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Class 10 Mathematics (Basic)

5 Sample Papers

Board Exam 2027 | CBSE 2026-27 syllabus | Latest 5-section paper pattern

5 full-length practice papers of 80 marks each (3 hours, 38 questions), put together from CBSE board questions of 2023 to 2026.

Paper	Difficulty
Paper 1	Easy
Paper 2	Moderate to Hard
Paper 3	Moderate to Hard
Paper 4	Moderate to Hard
Paper 5	Hard

Sample Paper 1

CLASS 10 | MATHEMATICS (BASIC)

Board Examination Practice Paper (2026-27 pattern) | Level: Easy

Maximum Marks: 80

Time Allowed: 3 Hours

GENERAL INSTRUCTIONS

1. This question paper contains **38** questions. **All questions are compulsory.**
2. This question paper is divided into **five** sections - Section A, B, C, D and E.
3. **Section A:** Q. Nos. 1 to 18 are Multiple Choice Questions (MCQs) and Q. Nos. 19 and 20 are Assertion-Reason based questions of **1 mark** each.
4. **Section B:** Q. Nos. 21 to 25 are Very Short Answer (VSA) type questions carrying **2 marks** each.
5. **Section C:** Q. Nos. 26 to 31 are Short Answer (SA) type questions carrying **3 marks** each.
6. **Section D:** Q. Nos. 32 to 35 are Long Answer (LA) type questions carrying **5 marks** each.
7. **Section E:** Q. Nos. 36 to 38 are case study based questions carrying **4 marks** each. Internal choice is provided in the 2-mark sub-part of each case study.
8. There is no overall choice. However, an internal choice has been provided in 2 questions in Section B (Q23, Q24), 2 in Section C (Q27, Q28), 2 in Section D (Q32, Q33) and 3 in Section E (Q36, Q37, Q38).
9. Draw neat diagrams wherever required. Take **$\pi = 22/7$** wherever required, if not stated.
10. Use of calculator is **NOT** allowed. Marks for each question are shown in square brackets on the right.

SECTION A - Multiple Choice and Assertion-Reason Questions (20 Marks)

1. HCF (132, 77) is : **[1]**

(A) 11 **(B)** 77

(C) 22 **(D)** 44

2. The value of k for which sum of the zeroes of the polynomial $p(x) = 3x^2 - kx + 6$ is 2, is **[1]**

(A) 2 **(B)** -6

(C) -2 **(D)** 6

3. The value of 'k' for which the pair of linear equations $x + y - 4 = 0$ and $2x + ky - 8 = 0$ has infinitely many solutions, is **[1]**

(A) $k \neq 2$ **(B)** $k \neq -2$

(C) $k = 2$ **(D)** $k = -2$

4. In an A.P., if $d = -4$ and $a_7 = 4$, then the first term 'a' is equal to **[1]**

(A) 6 **(B)** 7

(C) 20 **(D)** 28

5. If -26, *x*, 2 are in A.P., then the value of *x* is **[1]**

(A) 14 **(B)** -13

(C) -12 **(D)** -14

6. The sides of two similar triangles are in the ratio 4 : 7. The ratio of their perimeters is **[1]**

(A) 4 : 7 **(B)** 12 : 21

(C) 16 : 49 **(D)** 7 : 4

7. Which of the following is *not* the criterion for similarity of triangles ? **[1]**

(A) AAA **(B)** SSS

(C) SAS **(D)** RHS

8. For a point (3, -5), the value of (abscissa - ordinate) is : **[1]**

(A) - 8 **(B)** - 2

(C) 2 **(D)** 8

9. The distance of a point P(3, -7) from y-axis is : **[1]**

(A) 3 **(B)** 7

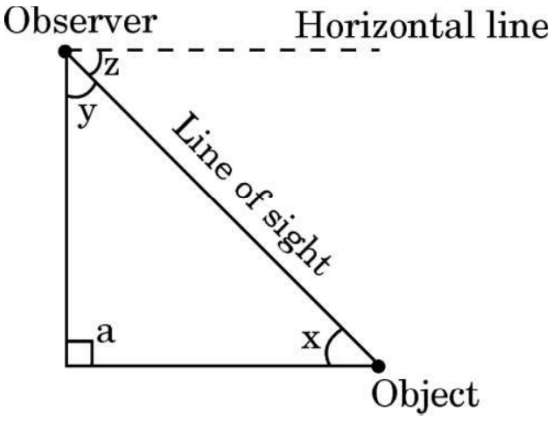
(C) - 7 **(D)** $\sqrt{58}$

10. The angle subtended by a tower of height 200 metres at a point 200 metres from the base is **[1]**

(A) 30° **(B)** 45°

(C) 60° **(D)** 0°

11. In the given figure, which of the following angles represents the angle of depression ? **[1]**



(A) x **(B)** y

(C) z **(D)** a

12. The distance between two parallel tangents of a circle of diameter 7 cm is : **[1]**

(A) 7 cm **(B)** 14 cm

(C) $\frac{7}{2}$ cm **(D)** 28 cm

13. A line intersecting a circle in two distinct points is called a **[1]**

(A) secant **(B)** chord

(C) diameter **(D)** tangent

14. The ratio of the area of a quadrant of a circle to the area of the same circle is : **[1]**

(A) 1 : 2 **(B)** 2 : 1

(C) 1 : 4 **(D)** 4 : 1

15. The volume of a cone of radius 'r' and height '3r' is : **[1]**

(A) $\frac{1}{3}\pi r^3$ **(B)** $3\pi r^3$

(C) $9\pi r^3$ **(D)** πr^3

16. The total surface area of a solid hemisphere of radius 7 cm is : **[1]**

(A) 98 π cm^2 **(B)** 147 π cm^2

(C) 196 π cm^2 **(D)** 228 $\frac{2}{3}$ π cm^2

17. A die is thrown once. Find the probability of getting a number less than 7. **[1]**

(A) $\frac{5}{6}$ **(B)** 1

(C) $\frac{1}{6}$ **(D)** 0

18. If probability of winning a game is p, then probability of losing the game is : **[1]**

(A) 1 + p **(B)** -p

(C) p - 1 **(D)** 1 - p

Assertion-Reason based questions. *Directions* : In Q. Nos. 19 and 20, a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

19. **Assertion (A) :** The point (0, 4) lies on y – axis. **[1]**

Reason (R) : The x-coordinate of a point, lying on y – axis, is zero.

20. **Assertion (A) :** For an acute angle θ, cos θ is always less than 1. **[1]**

Reason (R) : In a right-angled triangle, hypotenuse is the longest side and cos θ = $\frac{\text{Base}}{\text{Hypotenuse}}$

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SECTION B - Very Short Answer Questions (10 Marks)

21. Find the H.C.F. and L.C.M. of 408 and 312. **[2]**

22. Evaluate : $2(\sin^2 45° + \cot^2 30°) - 6(\cos^2 45° - \tan^2 30°)$ **[2]**

23. (a) If sec A = √2 and tan B = √3, then find the value of 2 sin A cos B. **[2]**

OR

(b) Evaluate : $\frac{4\cos^3 60° + \text{cosec } 30°}{\tan^2 30°}$

24. (a) Evaluate : $\frac{5\sin^2 45° - 3\tan^2 30°}{2\sec^2 30°}$ **[2]**

OR

(b) For A = 60° and B = 30°, verify that $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$.

25. From a circular sheet of radius 70 cm, a quadrant is cut. Find the area of the remaining sheet. **[2]**

SECTION C - Short Answer Questions (18 Marks)

26. Prove that $\sqrt{2}$ is an irrational number. **[3]**

27. (a) In an A.P., it is given that a = 2, d = 8 and S_n = 90. Find the value of n. **[3]**

OR

(b) How many 4-digit numbers are divisible by 7 ?

28. (a) The 4th and 10th term of an A.P. are 13 and 25 respectively. Find its 24th term. **[3]**

OR

(b) From 232 to 540, find the number of multiples of 3.

29. Show that the points A(1, 7), B(4, 2) C(-1, -1) and D(-4, 4) are vertices of the square ABCD. **[3]**

30. Prove that : $\tan^2\theta + \cot^2\theta + 2 = \sec^2\theta \operatorname{cosec}^2\theta$. **[3]**

31. Prove that the tangents drawn from an external point to a circle are equal in length. **[3]**

SECTION D - Long Answer Questions (20 Marks)

32. (a) The sum of reciprocals of Roohi's age (in years) 3 years ago and 5 years hence from now is $\frac{1}{3}$. Find her present age. **[5]**

OR

(b) A train travels 360 km at a uniform speed. If the speed had been 5 km/hr more, it would have taken 1 hour less for the same journey. Find the speed of the train.

33. (a) The difference of the squares of two positive numbers is 180. The square of the smaller number is 8 times the greater number. Find the two numbers. **[5]**

OR

(b) Find the value(s) of k for which the equation $2x^2 + kx + 3 = 0$ has real and equal roots. Hence, find the roots of the equations so obtained.

34. A heap of rice is in the form of a cone of base diameter 24 m and height $\frac{7}{2}$ m. Find the volume of rice. How much canvas cloth is required to just cover the heap ? **[5]**

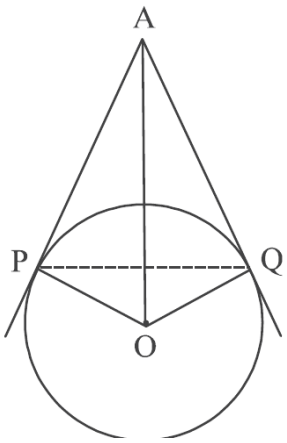
35. Find mean and mode of the following data : **[5]**

Class	0-15	15-30	30-45	45-60	60-75	75-90	90-105
Frequency	4	6	8	10	12	7	3

SECTION E - Case Study Based Questions (12 Marks)

36. Read the case study given below and answer the questions that follow : **[4]**

The picture given below shows a circular mirror hanging on the wall with a cord. The diagram represents the mirror as a circle with centre O. AP and AQ are tangents to the circle at P and Q respectively such that AP = 30 cm and ∠PAQ = 60°.



(36.1) Find the length PQ. **[1]**

(36.2) Find m ∠POQ. **[1]**

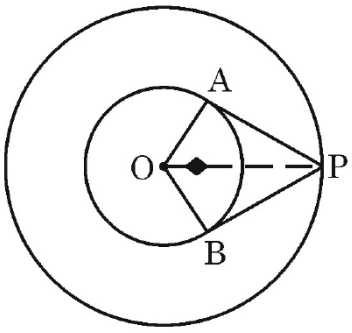
(36.3) (a) Find the length OA. **[2]**

OR

(36.3) (b) Find the radius of the mirror. **[2]**

37. Read the case study given below and answer the questions that follow : **[4]**

In a circular museum hall of radius 14 m, some statues are displayed. Statues are kept inside the inner concentric circle of radius 7 m. One such statue lying in sector OAB, is fenced along line segments OA, AP, PB and BO where P is a point on outer circle.
Based on above information, answer the following questions :



(37.1) Find m∠AOP. **[1]**

(37.2) Prove that ΔOAP ≅ ΔOBP. **[1]**

(37.3) (a) Find the length of fencing required to protect the statue. (Take √3 = 1.73) **[2]**

OR

(37.3) (b) Find area of quadrilateral OAPB. (Take √3 = 1.73) **[2]**

38. Read the case study given below and answer the questions that follow : **[4]**

Gurpreet is very fond of doing research on plants. She collected some leaves from different plants and measured their lengths in mm.
The data obtained is represented in the following table :

Length (in mm) :	70-80	80-90	90-100	100-110	110-120	120-130	130-140
Number of leaves :	3	5	9	12	5	4	2

(38.1) Write the median class of the data. **[1]**

(38.2) How many leaves are of length equal to or more than 10 cm ? **[1]**

(38.3) (a) Find median of the data. **[2]**

OR

(38.3) (b) Write the modal class and find the mode of the data. **[2]**

- End of Question Paper -

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Sample Paper 2

CLASS 10 | MATHEMATICS (BASIC)

Board Examination Practice Paper (2026-27 pattern) | Level: Moderate to Hard

Maximum Marks: 80

Time Allowed: 3 Hours

GENERAL INSTRUCTIONS

1. This question paper contains **38** questions. **All questions are compulsory.**
2. This question paper is divided into **five** sections - Section A, B, C, D and E.
3. **Section A:** Q. Nos. 1 to 18 are Multiple Choice Questions (MCQs) and Q. Nos. 19 and 20 are Assertion-Reason based questions of **1 mark** each.
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7. **Section E:** Q. Nos. 36 to 38 are case study based questions carrying **4 marks** each. Internal choice is provided in the 2-mark sub-part of each case study.
8. There is no overall choice. However, an internal choice has been provided in 2 questions in Section B (Q21, Q22), 2 in Section C (Q29, Q30), 2 in Section D (Q32, Q34) and 3 in Section E (Q36, Q37, Q38).
9. Draw neat diagrams wherever required. Take **π = 22/7** wherever required, if not stated.
10. Use of calculator is **NOT** allowed. Marks for each question are shown in square brackets on the right.

SECTION A - Multiple Choice and Assertion-Reason Questions (20 Marks)

1. The HCF of smallest 2 – digit number and the smallest composite number is : **[1]**

(A) 2 **(B)** 20

(C) 40 **(D)** 4

2. If the HCF of two positive integers a and b is 1, then their LCM is : **[1]**

(A) a + b **(B)** a

(C) b **(D)** ab

3. 17 × 11 × 13 + 11 is **[1]**

(A) a prime number. **(B)** multiple of 17.

(C) a composite number. **(D)** an odd number.

4. The sum and product of zeroes of the polynomial $p(x) = 3x^2 - 5x + 2$ are **[1]**

(A) $\frac{5}{3}, \frac{2}{3}$ **(B)** $\frac{-5}{3}, \frac{2}{3}$

(C) $1, \frac{2}{3}$ **(D)** $\frac{-5}{3}, \frac{-2}{3}$

5. The discriminant of the quadratic equation $2x^2 - 3x - 5 = 0$ is : **[1]**

(A) -31 **(B)** 49

(C) 7 **(D)** $\sqrt{(-31)}$

6. If the roots of quadratic equation $4x^2 - 5x + k = 0$ are real and equal, then value of k is : **[1]**

(A) $\frac{5}{4}$ **(B)** $\frac{25}{16}$

(C) $-\frac{5}{4}$ **(D)** $-\frac{25}{16}$

7. The seventh term of an A.P. whose first term is 28 and common difference -4, is **[1]**

(A) 0 **(B)** 4

(C) 52 **(D)** 56

8. If $\frac{-20}{9}, \frac{-2}{9}, \frac{16}{9}, \ldots$ are in A.P., then next term of the sequence is **[1]**

(A) $\frac{32}{9}$ **(B)** $\frac{46}{9}$

(C) $\frac{2}{9}$ **(D)** $\frac{34}{9}$

9. The distance between the points (2, -3) and (-2, 3) is : **[1]**

(A) $2\sqrt{13}$ units **(B)** 5 units

(C) $13\sqrt{2}$ units **(D)** 10 units

10. The mid-point of a line segment divides the line segment in the ratio : **[1]**

(A) 1 : 2 **(B)** 2 : 1

(C) 1 : 1 **(D)** $\frac{1}{2}$: 2

11. Which of the following statements is *false* ? **[1]**

(A) $\tan 45° = \cot 45°$ **(B)** $\sin 90° = \tan 45°$

(C) $\sin 30° = \cos 30°$ **(D)** $\sin 45° = \cos 45°$

12. The value of $(\frac{1}{\sec^2 A} + \frac{1}{\text{cosec}^2 A})$ is : **[1]**

(A) more than 1 **(B)** 1

(C) 0 **(D)** -1

13. If the distance of a tangent to a circle from its centre is 4 cm, then the length of diameter of the circle is : **[1]**

(A) 2 cm **(B)** 4 cm

(C) 8 cm **(D)** 16 cm

14. PQ is tangent to a circle at a point P on the circle. The number of tangents which can be drawn to the circle parallel to PQ, is **[1]**

(A) 2 **(B)** 1

(C) many **(D)** zero

15. The total surface area of a solid cone of radius 7 cm and slant height 25 cm, is **[1]**

(A) 724 cm^2 **(B)** 704 cm^2

(C) 550 cm^2 **(D)** 616 cm^2

16. The mean and median of a statistical data are 21 and 23 respectively. The mode of the data is : **[1]**

(A) 27 **(B)** 22

(C) 17 **(D)** 23

17. If P(A) denotes the probability of an event A, then **[1]**

(A) $P(A) < 0$ **(B)** $P(A) > 1$

(C) $0 \leq P(A) \leq 1$ **(D)** $-1 \leq P(A) \leq 1$

18. The probability of getting sum greater than 10, when two dice are rolled together, is **[1]**

(A) $\frac{1}{9}$ **(B)** $\frac{1}{18}$

(C) $\frac{1}{12}$ **(D)** 1

Assertion-Reason based questions. *Directions* : In Q. Nos. 19 and 20, a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

19. **Assertion (A) :** A tangent to a circle is perpendicular to the radius through the point of contact. **[1]**

Reason (R) : The lengths of tangents drawn from the external point to a circle are equal.

20. **Assertion (A) :** Median of a data is the value of $\frac{N}{2}$, where N represents sum of all frequencies. **[1]**

Reason (R) : Median divides the whole distribution in two equal parts.

SECTION B - Very Short Answer Questions (10 Marks)

21. (a) The sum of two natural numbers is 70 and their difference is 10. Find the natural numbers. **[2]**

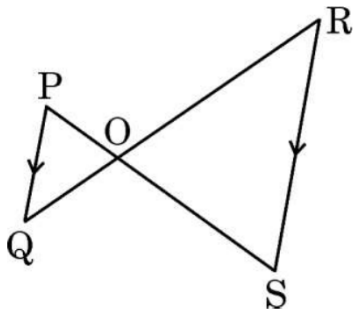
OR

(b) Solve for x and y :

$x - 3y = 7$

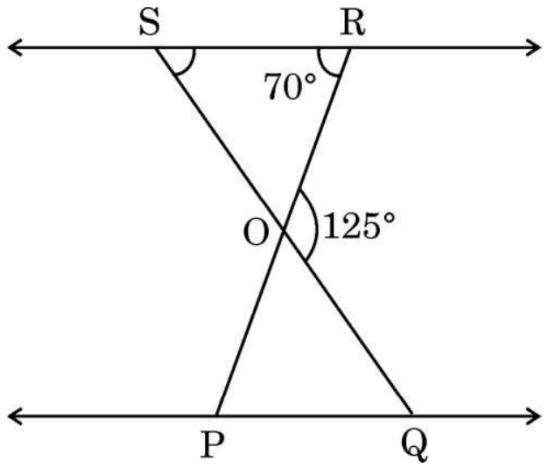
$3x - 3y = 5$

22. (a) In the given figure, if PQ ∥ RS, then prove that Δ POQ ~ Δ SOR. **[2]**



OR

(b) In the given figure, Δ OSR ~ Δ OQP, ∠ ROQ = 125° and ∠ ORS = 70°. Find the measures of ∠ OSR and ∠ OQP.



23. In a ΔABC, ∠A = 90°. If tan C = √3, then find the value of sin B + cos C - $\cos^2$ B. **[2]**

24. 15 defective pens are accidentally mixed with 145 good ones. One pen is taken out at random from this lot. Determine the probability that the pen taken out is a good one. **[2]**

25. A bag contains 4 red, 5 white and some yellow balls. If probability of drawing a red ball at random is $\frac{1}{5}$, then find the probability of drawing a yellow ball at random. **[2]**

SECTION C - Short Answer Questions **(18 Marks)**

26. Prove that 7 - 3√5 is an irrational number, given that √5 is an irrational number. **[3]**

27. If α, β are zeroes of the quadratic polynomial x^2 + 3x + 2, find a quadratic polynomial whose zeroes are α + 1, β + 1. **[3]**

28. Determine a quadratic polynomial, sum and product of whose zeroes are -10 and 24, respectively. Also, determine the zeroes of the polynomial so obtained. **[3]**

29. (a) How many terms of the A.P. 3, 5, 7, 9, ... must be added to get the sum 80 ? **[3]**

OR

(b) Find three consecutive terms in A.P. whose sum is 21 and their product is 231.

30. (a) Prove that **[3]**

$\sec\theta\,(1 - \sin\theta)\,(\sec\theta + \tan\theta) = 1$

OR

(b) Prove that

$$\frac{1 + \sec\theta}{\sec\theta} = \frac{\sin^2\theta}{1 - \cos\theta}$$

31. Two dice are tossed simultaneously. Find the probability of getting **[3]**

(a) an even number on both the dice.

(b) the sum of two numbers more than 9.

SECTION D - Long Answer Questions (20 Marks)

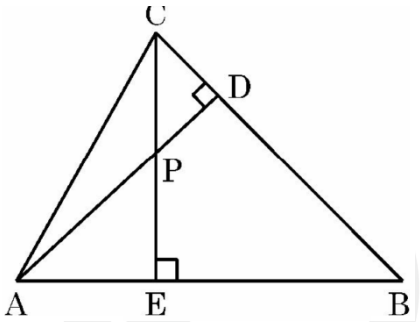
32. (a) In the given figure, altitudes CE and AD of ΔABC intersect each other at the point P. **[5]**

Show that

(i) ΔAEP ~ ΔCDP

(ii) ΔABD ~ ΔCBE

(iii) ΔAEP ~ ΔADB



OR

(b) AD and PM are medians of triangles ABC and PQR, respectively, where ΔABC ~ ΔPQR.

Prove that $\frac{AB}{PQ} = \frac{AD}{PM}$.

33. State the converse of "Basic Proportionality Theorem" and use it to prove the following : **[5]**

Line segment joining mid-points of any two sides of a triangle is parallel to the third side.

34. (a) The shadow of a tower standing on a level ground is found to be 40 m longer when the Sun's altitude is 30° than when it was 60°. Find the height of the tower and the length of original shadow. (use $\sqrt{3} = 1.73$) **[5]**

OR

(b) The angles of depression of the top and the bottom of an 8 m tall building from the top of a multi-storeyed building are 30° and 45° respectively. Find the height of the multi-storeyed building and the distance between the two buildings. (use $\sqrt{3} = 1.73$)

35. A chord of a circle of radius 14 cm subtends an angle of 90° at the centre. Find the area of the corresponding minor and major segments of the circle. **[5]**

SECTION E - Case Study Based Questions (12 Marks)

36. Read the case study given below and answer the questions that follow : **[4]**

> Saving money is a good habit and it should be inculcated in children right from the beginning. Rehan's mother brought a piggy bank for Rehan and puts one ₹ 5 coin of her savings in the piggy bank on the first day. She increases his savings by one ₹ 5 coin daily.

(36.1) How many coins were added to the piggy bank on 8th day ? **[1]**

(36.2) How much money will be there in the piggy bank after 8 days ? **[1]**

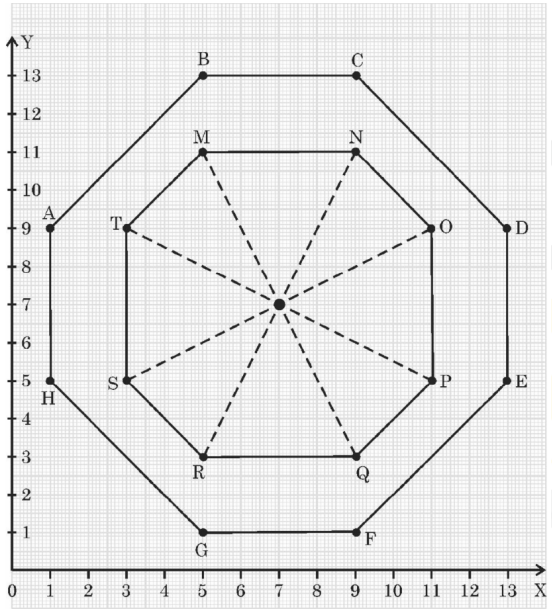
(36.3) (a) If the piggy bank can hold one hundred twenty ₹ 5 coins in all, find the number of days she can contribute to put ₹ 5 coins into it. **[2]**

OR

(36.3) (b) Find the total money saved, when the piggy bank is full. **[2]**

37. Read the case study given below and answer the questions that follow : **[4]**

> The top of a table is octagonal in shape.



(37.1) Write the coordinates of A and B. **[1]**

(37.2) Write the coordinates of the mid-point of line segment joining C and D. **[1]**

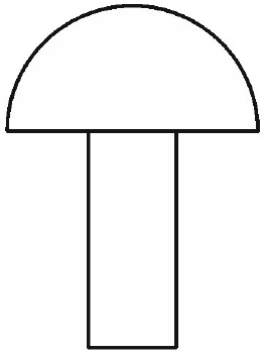
(37.3) (a) Find the distance between M and Q. **[2]**

OR

(37.3) (b) Find the coordinates of the point which divides the line segment joining M and N in the ratio 1:3 internally. **[2]**

38. Read the case study given below and answer the questions that follow : **[4]**

There are many varieties of mushrooms available in the world. One such mushroom 'Amanita muscaria' has a upper part which is like red cap (hemispherical) and lower part is like white stem (cylinderical).
The hemispherical cap's radius = 3 cm and cylindrical stem is 2 cm high with diameter 1.4 cm. Considering mushroom a solid object, answer the following questions :



(38.1) What is the total height of a mushroom ? **[1]**

(38.2) Find the volume of the stem. **[1]**

(38.3) (a) Determine the volume of 7 such mushrooms. **[2]**

OR

(38.3) (b) Find the total surface area of 7 such mushrooms. **[2]**

- End of Question Paper -

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Sample Paper 3

CLASS 10 | MATHEMATICS (BASIC)

Board Examination Practice Paper (2026-27 pattern) | Level: Moderate to Hard

Maximum Marks: 80

Time Allowed: 3 Hours

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7. **Section E:** Q. Nos. 36 to 38 are case study based questions carrying **4 marks** each. Internal choice is provided in the 2-mark sub-part of each case study.
8. There is no overall choice. However, an internal choice has been provided in 2 questions in Section B (Q21, Q24), 2 in Section C (Q27, Q28), 2 in Section D (Q32, Q34) and 3 in Section E (Q36, Q37, Q38).
9. Draw neat diagrams wherever required. Take **π = 22/7** wherever required, if not stated.
10. Use of calculator is **NOT** allowed. Marks for each question are shown in square brackets on the right.

SECTION A - Multiple Choice and Assertion-Reason Questions (20 Marks)

1. The prime factorisation of natural number 288 is **[1]**

(A) $2^4 \times 3^3$ **(B)** $2^4 \times 3^2$

(C) $2^5 \times 3^2$ **(D)** $2^5 \times 3^1$

2. $(2 + \sqrt{2})^2$ is : **[1]**

(A) a rational number **(B)** an irrational number

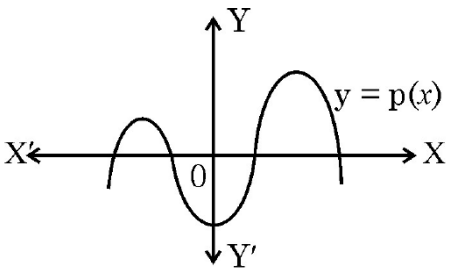
(C) an integer **(D)** a natural number

3. A quadratic polynomial, the sum of whose zeroes is -5 and their product is 6, is **[1]**

(A) $x^2 + 5x + 6$ **(B)** $x^2 - 5x + 6$

(C) $x^2 - 5x - 6$ **(D)** $-x^2 + 5x + 6$

4. The graph of a polynomial $p(x)$ is shown here. The number of zeroes of the polynomial $p(x)$ is **[1]**



(A) 5 **(B)** 1

(C) 0 **(D)** 4

5. If the zeroes of the polynomial $p(x) = 2x^2 - 7x + 6$ are α and β, then the value of $\frac{1}{\alpha} + \frac{1}{\beta}$ is **[1]**

(A) $\frac{7}{2}$ **(B)** $\frac{6}{7}$

(C) $\frac{-7}{6}$ **(D)** $\frac{7}{6}$

6. The roots of the quadratic equation $x^2 + 9 = 0$ are **[1]**

(A) real and equal **(B)** not real

(C) real and negative of each other **(D)** rational numbers

7. The mid-point of the line segment joining the points (-1, 3) and $(8, \frac{3}{2})$ is : **[1]**

(A) $(\frac{7}{2}, -\frac{3}{4})$ **(B)** $(\frac{7}{2}, \frac{9}{2})$

(C) $(\frac{9}{2}, -\frac{3}{4})$ **(D)** $(\frac{7}{2}, \frac{9}{4})$

8. For a point X(a, b) where (b > a > 0), the value of its [distance from x-axis - distance from y-axis] is : **[1]**

(A) a - b **(B)** b - a

(C) $a^2 - b^2$ **(D)** $b^2 - a^2$

9. If sin A = $\frac{3}{5}$, then value of cot A is : **[1]**

(A) $\frac{3}{4}$ **(B)** $\frac{4}{3}$

(C) $\frac{4}{5}$ **(D)** $\frac{5}{4}$

10. If a pole 6 m high casts a shadow $2\sqrt{3}$ m long on the ground, then the sun's elevation is **[1]**

(A) 30° **(B)** 45°

(C) 60° **(D)** 90°

11. The height and radius of a right circular cone are 24 cm and 7 cm respectively. The slant height of the cone is : **[1]**

(A) 24 cm **(B)** 31 cm

(C) 26 cm **(D)** 25 cm

12. For which of the following solids is the lateral/curved surface area and total surface area the same ? **[1]**

(A) Cube **(B)** Cuboid

(C) Hemisphere **(D)** Sphere

13. If a certain variable x divides a statistical data arranged in order into two equal parts, then the value of x is called the : **[1]**

(A) mean **(B)** median

(C) mode **(D)** range

14. A card is drawn from a well-shuffled deck of 52 playing cards. The probability of getting a queen of spade is **[1]**

(A) $\frac{1}{26}$ **(B)** $\frac{1}{52}$

(C) 0 **(D)** $\frac{1}{4}$

15. On a throw of a die, if getting 6 is considered success then probability of losing the game is : **[1]**

(A) 0 **(B)** 1

(C) $\frac{1}{6}$ **(D)** $\frac{5}{6}$

16. In a random experiment of throwing a die, which of the following is a sure event ? **[1]**

(A) Getting a number between 1 and 6 **(B)** Getting an odd number < 7

(C) Getting an even number < 7 **(D)** Getting a natural number < 7

17. A bag contains 3 red, 4 white and 7 green balls. A ball is drawn at random. The probability that the ball drawn is *not* of red colour is : **[1]**

(A) $\frac{1}{11}$ **(B)** $\frac{3}{14}$

(C) $\frac{11}{14}$ **(D)** $\frac{3}{11}$

18. In an experiment of throwing a pair of dice, the probability of not getting a doublet is : **[1]**

(A) $\frac{1}{6}$ **(B)** $\frac{5}{6}$

(C) $\frac{1}{5}$ **(D)** $\frac{1}{30}$

Assertion-Reason based questions. *Directions* : In Q. Nos. 19 and 20, a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

19. **Assertion (A) :** For any two natural numbers a and b, the HCF of a and b is a factor of the LCM of a and b. **[1]**

Reason (R) : HCF of any two natural numbers divides both the numbers.

20. **Assertion (A) :** A line drawn parallel to any one side of a triangle intersects the other two sides in the same ratio. **[1]**

Reason (R) : Parallel lines cannot be drawn to any side of a triangle.

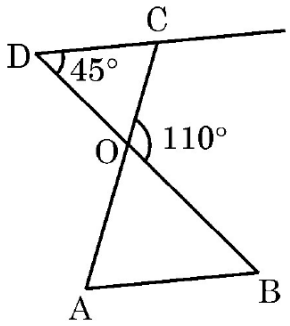
SECTION B - Very Short Answer Questions (10 Marks)

21. (a) Solve for x and y : x + y = 6, 2x - 3y = 4. **[2]**

OR

(b) Find out whether the following pair of linear equations are consistent or inconsistent : 5x - 3y = 11, -10x + 6y = 22

22. In the given figure, $\Delta ODC \sim \Delta OBA$. If $\angle BOC = 110°$, $\angle ODC = 45°$ and AB = 2 CD, then find (i) $m\angle OAB$ (ii) OB : OD. **[2]**



23. Evaluate : $\frac{3}{2}\tan^2 30° - 2\cos^2 90° - \frac{1}{2}\operatorname{cosec}^2 30°$ **[2]**

24. (a) Find the values of A and B ($0 \le A < 90°$, $0 \le B < 90°$), if $\tan(A + B) = 1$ and $\tan(A - B) = \frac{1}{\sqrt{3}}$. **[2]**

OR

(b) Prove that $\tan 45° = 1$ geometrically.

25. Two concentric circles are of radii 6 cm and 10 cm. Find the length of the chord of the larger circle which touches the smaller circle. **[2]**

SECTION C - Short Answer Questions (18 Marks)

26. Two alarm clocks ring their alarms at regular intervals of 20 minutes and 25 minutes respectively. If they first beep together at 12 noon, at what time will they beep again together next time ? **[3]**

27. (a) Find the zeroes of the polynomial $p(x) = 4x^2 - 8x + 3$ and verify the relationship between its zeroes and co-efficients. **[3]**

OR

(b) Form a polynomial whose zeroes are α^2 and β^2, where α and β are zeroes of the polynomial $p(x) = x^2 - 3\sqrt{2}x + 4$.

28. (a) Solve the following system of equations graphically : **[3]**
x + 3y = 6; 2x - 3y = 12

OR

(b) x and y are complementary angles such that x : y = 1 : 2. Express the given information as a system of linear equations in two variables and hence solve it.

29. Prove that : $(\cot\theta - \operatorname{cosec}\theta)^2 = \frac{1 - \cos\theta}{1 + \cos\theta}$. **[3]**

30. Prove the following trigonometric identity : **[3]**

$$\sqrt{\left(\frac{\text{cosec A} - 1}{\text{cosec A} + 1}\right)} = \sec A - \tan A$$

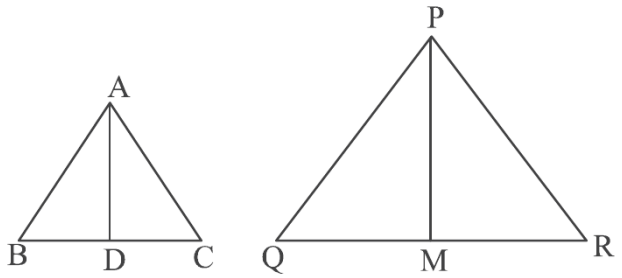
31. A car has two wipers which do not overlap. Each wiper has a blade of length 21 cm sweeping through an angle of 120°. Find the area cleaned at each sweep of the blades. **[3]**

SECTION D - Long Answer Questions (20 Marks)

32. (a) If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, then prove that other two sides are divided in the same ratio. **[5]**

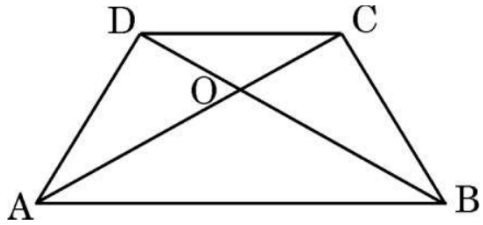
OR

(b) Sides AB and BC and median AD of a ΔABC are respectively proportional to sides PQ and PR and median PM of ΔPQR. Show that ΔABC ~ ΔPQR.



33. State "Basic Proportionality Theorem" and use it to prove the following : **[5]**

In a quadrilateral ABCD, diagonals AC and BD intersect each other at O such that $\frac{AO}{BO} = \frac{CO}{DO}$ as shown in the given figure. Prove that ABCD is a trapezium.



34. (a) A toy is in the form of a cone surmounted on a hemisphere. The cone and hemisphere have the same radii. The height of the conical part of the toy is equal to the diameter of its base. If the radius of the conical part is 5 cm, find the volume of the toy. **[5]**

OR

(b) A cubical block is surmounted by a hemisphere of radius 3.5 cm. What is the smallest possible length of the edge of the cube so that the hemisphere can totally lie on the cube ? Find the total surface area of the solid so formed.

35. The table given below shows the daily expenditure on food of 25 households in a locality : **[5]**

Daily expenditure (₹)	100 – 150	150 – 200	200 – 250	250 – 300	300 – 350
Number of household	4	5	12	2	2

Find the mean daily expenditure on food. Also, find the mode of the data.

SECTION E - Case Study Based Questions (12 Marks)

36. Read the case study given below and answer the questions that follow : **[4]**

Seema daily goes to a park to exercise on machines available there. When Seema spent 15 minutes on exercise bicycle and 30 minutes on double cross walker, she received a message of burning 435 calories on her fitness watch. When she spent 30 minutes on exercise bicycle and 40 minutes on double cross walker, she received a message of burning 690 calories.
To find the number of calories burned per minute on each machine, answer the following :

(36.1) Represent the above situation in terms of a pair of linear equations in two variables. **[1]**

(36.2) Show that the equations have unique solution. **[1]**

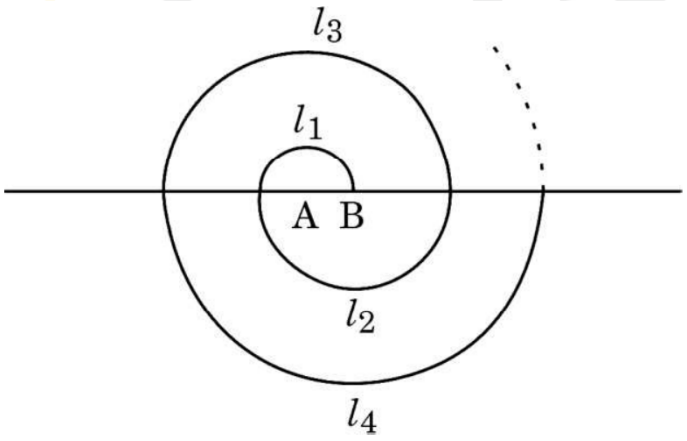
(36.3) (a) Solve both equations to find the values of the variables using elimination method. **[2]**

OR

(36.3) (b) Solve both equations to find the values of the variables using substitution method. **[2]**

37. Read the case study given below and answer the questions that follow : **[4]**

In a garden, saplings of rose flowers were planted at equal intervals to form a spiral pattern. The spiral is made up of successive semicircles, with centres alternatively at A and B, starting with centre at A, of radii 50 cm, 100 cm, 150 cm, ... as shown in the figure given below. Spiral 1 has 10 flowers, Spiral 2 has 20 flowers, Spiral 3 has 30 flowers and so on.



(37.1) What is the radius of the 13th spiral ? **[1]**

(37.2) If the radius of the nth spiral is 500 cm, find the value of n. **[1]**

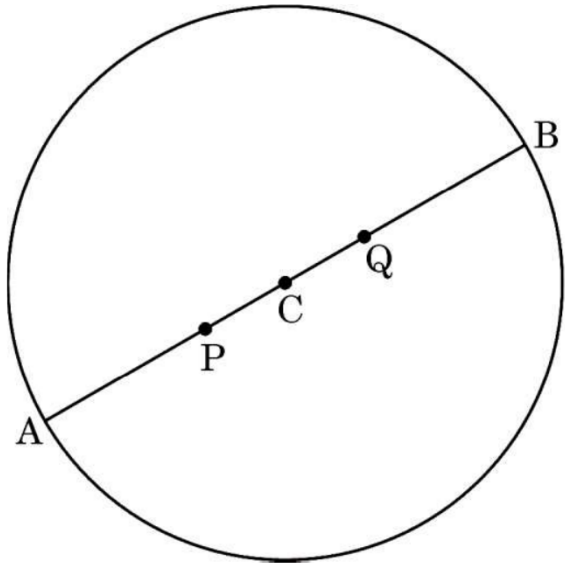
(37.3) (a) Find the total number of saplings till the 11th spiral. **[2]**

OR

(37.3) (b) Till which spiral, will there be a total of 450 saplings ? **[2]**

38. Read the case study given below and answer the questions that follow : **[4]**

In a society, there is a circular park having two gates. The gates are placed at points A(10, 20) and B(50, 50), as shown in the figure below. Two fountains are installed at points P and Q on AB such that AP = PQ = QB.



(38.1) Find the coordinates of the centre C. **[1]**

(38.2) Find the radius of the circular park. **[1]**

(38.3) (a) Find the coordinates of the point P. **[2]**

OR

(38.3) (b) Find the distance of the fountain at Q from gate A. **[2]**

- End of Question Paper -

Career Ka Dost
Ask before you decide

Sample Paper 4

CLASS 10 | MATHEMATICS (BASIC)

Board Examination Practice Paper (2026-27 pattern) | Level: Moderate to Hard

Maximum Marks: 80

Time Allowed: 3 Hours

GENERAL INSTRUCTIONS

1. This question paper contains **38** questions. **All questions are compulsory.**
2. This question paper is divided into **five** sections - Section A, B, C, D and E.
3. **Section A:** Q. Nos. 1 to 18 are Multiple Choice Questions (MCQs) and Q. Nos. 19 and 20 are Assertion-Reason based questions of **1 mark** each.
4. **Section B:** Q. Nos. 21 to 25 are Very Short Answer (VSA) type questions carrying **2 marks** each.
5. **Section C:** Q. Nos. 26 to 31 are Short Answer (SA) type questions carrying **3 marks** each.
6. **Section D:** Q. Nos. 32 to 35 are Long Answer (LA) type questions carrying **5 marks** each.
7. **Section E:** Q. Nos. 36 to 38 are case study based questions carrying **4 marks** each. Internal choice is provided in the 2-mark sub-part of each case study.
8. There is no overall choice. However, an internal choice has been provided in 2 questions in Section B (Q22, Q25), 2 in Section C (Q26, Q27), 2 in Section D (Q32, Q33) and 3 in Section E (Q36, Q37, Q38).
9. Draw neat diagrams wherever required. Take **π = 22/7** wherever required, if not stated.
10. Use of calculator is **NOT** allowed. Marks for each question are shown in square brackets on the right.

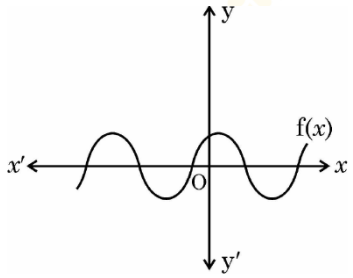
SECTION A - Multiple Choice and Assertion-Reason Questions (20 Marks)

1. LCM (850, 500) is : **[1]**

(A) 850 × 50 **(B)** 17 × 500

(C) $17 \times 5^2 \times 2^2$ **(D)** $17 \times 5^3 \times 2$

2. The graph of y = f(x) is shown in the figure for some polynomial f(x). **[1]**
The number of zeroes of f(x) is



(A) 5 **(B)** 6

(C) 4 **(D)** 8

3. The zeroes of the polynomial $3x^2 + 11x - 4$ are : **[1]**

(A) $\frac{1}{3}, 4$ **(B)** $\frac{-1}{3}, -4$

(C) $\frac{1}{3}, -4$ **(D)** $\frac{-1}{3}, 4$

4. The graph of a pair of linear equations $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$ in two variables x and y represents parallel lines, if **[1]**

(A) $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ **(B)** $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

(C) $\frac{a_1}{a_2} \neq \frac{b_1}{b_2} = \frac{c_1}{c_2}$ **(D)** $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

5. n^{th} term of the A.P. : $\frac{-1}{3}, \frac{4}{3}$, 3, ... is **[1]**

(A) (5n - 9)/3 **(B)** (5n - 6)/3

(C) (3n - 4)/3 **(D)** (3n + 2)/3

6. Which of the following statements is not always true ? **[1]**

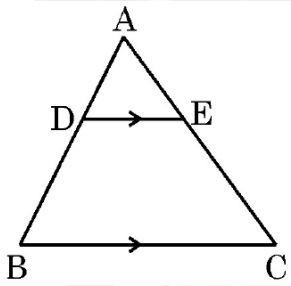
(A) Two circles are similar.

(B) Two isosceles right triangles are similar.

(C) Two rectangles are similar.

(D) Two equilateral triangles are similar.

7. In the given figure, DE ∥ BC. **[1]**
If AD : AB = 1 : 3 and AE = 2.5 cm, then AC equals



(A) 7.5 cm **(B)** 5 cm

(C) 10 cm **(D)** 2.5 cm

8. The diameter of a circle is of length 6 cm. If one end of the diameter is (-4, 0), the other end on x-axis is at : **[1]**

(A) (0, 2) **(B)** (6, 0)

(C) (2, 0) **(D)** (4, 0)

9. $\frac{1 + \tan^2 A}{1 + \cot^2 A}$ is equal to **[1]**

(A) $\sec^2 A$ **(B)** -1

(C) $\cot^2 A$ **(D)** $\tan^2 A$

10. If $\sin \theta = \frac{1}{3}$, then sec θ is equal to : **[1]**

(A) $(2\sqrt{2})/3$ **(B)** $3/(2\sqrt{2})$

(C) 3 **(D)** $1/\sqrt{3}$

11. For what value of θ, $\sin^2\theta + \sin \theta + \cos^2\theta$ is equal to 2 ? **[1]**

(A) 45° **(B)** 0°

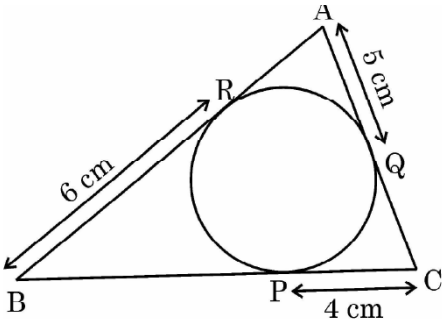
(C) 90° **(D)** 30°

12. If value of cot θ is $\sqrt{5}$, then sin θ equals **[1]**

(A) $\frac{1}{\sqrt{6}}$ **(B)** $\sqrt{6}$

(C) $\frac{\sqrt{5}}{6}$ **(D)** $\frac{1}{2}$

13. In the given figure, the perimeter of ΔABC is : **[1]**



(A) 30 cm **(B)** 15 cm

(C) 45 cm **(D)** 60 cm

14. The length of a pendulum is 70 cm and it describes an arc of length 88 cm when swings. The angle subtended by the arc at the centre is **[1]**

(A) 36° **(B)** 70°

(C) 72° **(D)** 80°

15. The class mark of the median class of the following data is : **[1]**

Class Interval	10 – 25	25 – 40	40 – 55	55 – 70	70 – 85	85 – 100
Frequency	2	3	7	6	6	6

(A) 40 **(B)** 55

(C) 47.5 **(D)** 62.5

16. The following distribution shows the number of runs scored by some batsmen in test matches : **[1]**

Runs Scored	3000 – 4000	4000 – 5000	5000 – 6000	6000 – 7000
Number of Batsmen	5	10	9	8

The lower limit of the modal class is :

(A) 3000 **(B)** 4000

(C) 5000 **(D)** 6000

17. Two dice are rolled together. The probability of getting a doublet is : **[1]**

(A) $\frac{2}{36}$ **(B)** $\frac{1}{36}$

(C) $\frac{1}{6}$ **(D)** $\frac{5}{6}$

18. A bag contains some red and some white balls. A ball is drawn at random from the bag. If the probability of getting a red ball is $\frac{2}{7}$, then the probability of getting a white ball is **[1]**

(A) $\frac{1}{14}$ **(B)** $\frac{5}{7}$

(C) $\frac{1}{7}$ **(D)** $\frac{2}{7}$

Assertion-Reason based questions. *Directions* : In Q. Nos. 19 and 20, a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).
(C) Assertion (A) is true, but Reason (R) is false.
(D) Assertion (A) is false, but Reason (R) is true.

19. **Assertion (A) :** Zeroes of a polynomial $p(x) = x^2 - 2x - 3$ are -1 and 3. **[1]**
Reason (R) : The graph of polynomial $p(x) = x^2 - 2x - 3$ intersects x-axis at (-1, 0) and (3, 0).

20. **Assertion (A) :** The system of linear equations $3x + 5y - 4 = 0$ and $15x + 25y - 25 = 0$ is inconsistent. **[1]**
Reason (R) : The pair of linear equations $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ is inconsistent if $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$.

SECTION B - Very Short Answer Questions (10 Marks)

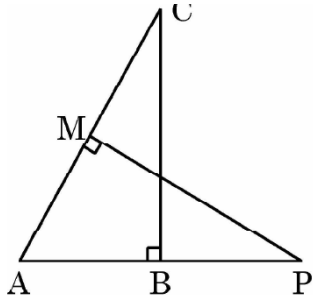
21. Find the HCF of 84 and 144 by prime factorisation method. **[2]**

22. (a) Solve the following pair of linear equations for x and y algebraically : **[2]**
$x + 2y = 9$ and $y - 2x = 2$

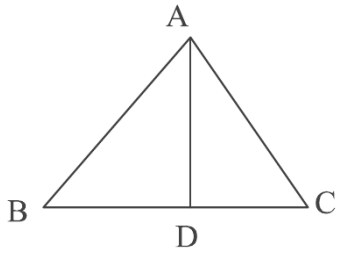
OR

(b) Check whether the point (-4, 3) lies on both the lines represented by the linear equations $x + y + 1 = 0$ and $x - y = 1$.

23. In the given figure, ABC and AMP are two right triangles, right angled at B and M, respectively. Prove that $\Delta ABC \sim \Delta AMP$. **[2]**



24. D is a point on the side BC of ΔABC such that ∠ADC = ∠BAC. Show that $AC^2 = BC \times DC$. **[2]**



25. (a) Find the coordinates of the point which divides the line segment joining the points (7, -1) and (-3, 4) internally in the ratio 2 : 3. **[2]**

OR

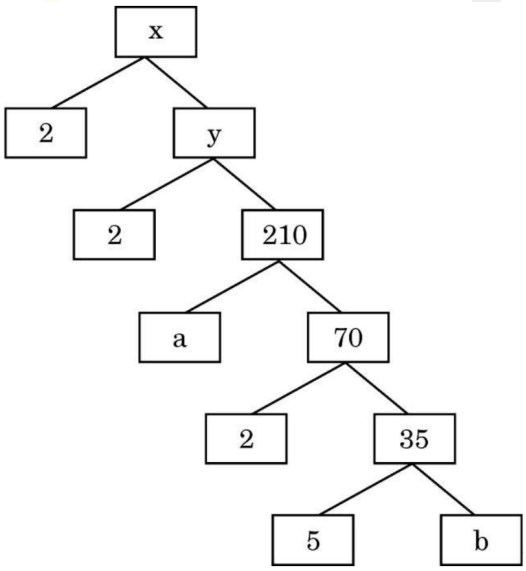
(b) Find the value(s) of y for which the distance between the points A(3, -1) and B(11, y) is 10 units.

SECTION C - Short Answer Questions (18 Marks)

26. (a) Prove that $\sqrt{3}$ is an irrational number. **[3]**

OR

(b) The factor tree of a number x is shown below :
Find the values of x, y, a and b. Hence, write the product of the prime factors of the number x so obtained.



27. (a) If α, β are zeroes of the polynomial $p(x) = 5x^2 - 7x - 3$, then form a quadratic polynomial whose zeroes are $\frac{2}{\alpha}$ and $\frac{2}{\beta}$. **[3]**

OR

(b) Find the zeroes of the polynomial $p(x) = 3x^2 + 7x - 20$ and verify the relationship between its zeroes and the coefficients.

28. Find the co-ordinates of the points of trisection of the line segment joining the points (-2, 2) and (7, -4). **[3]**

29. Prove that $\frac{\tan\theta}{1 - \cot\theta} + \frac{\cot\theta}{1 - \tan\theta} = 1 + \sec\theta \operatorname{cosec}\theta$ **[3]**

30. Prove that a rectangle circumscribing a circle is a square. **[3]**

31. A lot consists of 200 pens of which 180 are good and the rest are defective. A customer will buy a pen if it is not defective. The shopkeeper draws a pen at random and gives it to the customer. What is the probability that the customer will not buy it ? Another lot of 100 pens containing 80 good pens is mixed with the previous lot of 200 pens. The shopkeeper now draws one pen at random from the entire lot and gives it to the customer. What is the probability that the customer will buy the pen ? **[3]**

SECTION D - Long Answer Questions **(20 Marks)**

32. (a) The diagonal of a rectangular field is 60 m more than the shorter side. If the longer side is 80 m more than the shorter side, find the length of the sides of the field. **[5]**

OR

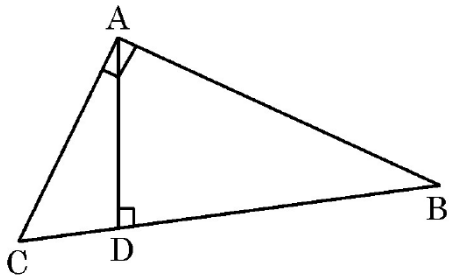
(b) The sum of the ages of a father and his son is 45 years. Five years ago, the product of their ages (in years) was 124. Determine their present age.

33. (a) In the given figure, ΔABC is right angled triangle with ∠A = 90°. AD is perpendicular to BC. Prove that : **[5]**

(i) ΔDBA ~ ΔDAC

(ii) $DA^2 = DB \times DC$

(iii) Find the area of ΔABC when DB = 9 cm and DC = 16 cm.



OR

(b) If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, then prove that the other two sides are divided in the same ratio.

34. A TV tower stands vertically on a bank of a canal. From a point on the other bank exactly opposite the tower, the angle of elevation of the top of the tower is 60°. From another point 20 m away from this point on the line joining this point to the foot of the tower, the angle of elevation of the top of the tower is 30°. **[5]**

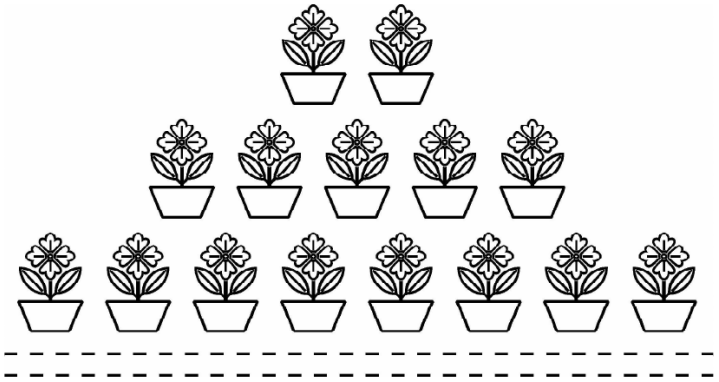
Find the height of the tower and the width of the canal. [Use √3 = 1.732]

35. A vessel is in the form of a hemispherical bowl surmounted by a hollow cylinder of same diameter. The diameter of the hemispherical bowl is 14 cm and the total height of the vessel is 13 cm. Find the inner surface area of the vessel. Also, find the volume of the vessel. **[5]**

SECTION E - Case Study Based Questions (12 Marks)

36. Read the case study given below and answer the questions that follow : **[4]**

Aahana being a plant lover decides to convert her balcony into beautiful garden full of plants. She bought few plants with pots for her balcony. She placed the pots in such a way that number of pots in the first row is 2, second row is 5, third row is 8 and so on.



(36.1) Find the number of pots placed in the 10th row. **[1]**

(36.2) Find the difference in the number of pots placed in 5th row and 2nd row. **[1]**

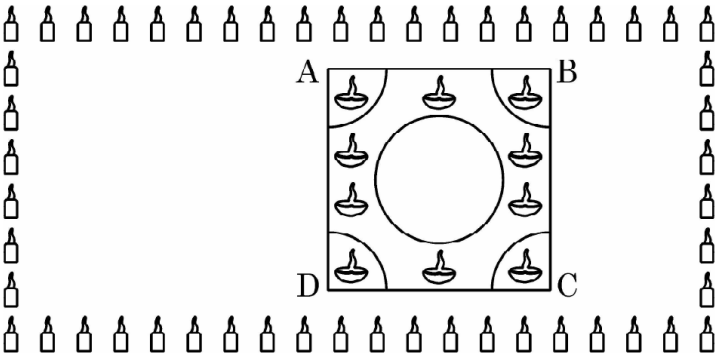
(36.3) (a) If Aahana wants to place 100 pots in total, then find the total number of rows formed in the arrangement. **[2]**

OR

(36.3) (b) If Aahana has sufficient space for 12 rows, then how many total number of pots are placed by her with the same arrangement ? **[2]**

37. Read the case study given below and answer the questions that follow : **[4]**

Interschool Rangoli Competition was organized by one of the reputed schools of Odissa. The theme of the Rangoli Competition was Diwali celebrations where students were supposed to make mathematical designs. Students from various schools participated and made beautiful Rangoli designs. One such design is given below.
Rangoli is in the shape of square marked as ABCD, side of square being 40 cm. At each corner of a square, a quadrant of circle of radius 10 cm is drawn (in which diyas are kept). Also a circle of diameter 20 cm is drawn inside the square.



(37.1) What is the area of square ABCD ? **[1]**

(37.2) Find the area of the circle. **[1]**

(37.3) (a) If the circle and the four quadrants are cut off from the square ABCD and removed, then find the area of remaining portion of square ABCD. **[2]**

OR

(37.3) (b) Find the combined area of 4 quadrants and the circle, removed. **[2]**

38. Read the case study given below and answer the questions that follow : **[4]**

Blood group describes the type of blood a person has. It is a classification of blood based on the presence or absence of inherited antigenic substances on the surface of red blood cells. Blood types predict whether a serious reaction will occur in a blood transfusion.
In a sample of 50 people, 21 had type O blood, 22 had type A, 5 had type B and rest had type AB blood group.

(38.1) What is the probability that a person chosen at random had type O blood ? **[1]**

(38.2) What is the probability that a person chosen at random had type AB blood group ? **[1]**

(38.3) (a) What is the probability that a person chosen at random had neither type A nor type B blood group ? **[2]**

OR

(38.3) (b) What is the probability that person chosen at random had either type A or type B or type O blood group ? **[2]**

- End of Question Paper -

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Sample Paper 5

CLASS 10 | MATHEMATICS (BASIC)

Board Examination Practice Paper (2026-27 pattern) | Level: Hard

Maximum Marks: 80

Time Allowed: 3 Hours

GENERAL INSTRUCTIONS

1. This question paper contains **38** questions. **All questions are compulsory.**
2. This question paper is divided into **five** sections - Section A, B, C, D and E.
3. **Section A:** Q. Nos. 1 to 18 are Multiple Choice Questions (MCQs) and Q. Nos. 19 and 20 are Assertion-Reason based questions of **1 mark** each.
4. **Section B:** Q. Nos. 21 to 25 are Very Short Answer (VSA) type questions carrying **2 marks** each.
5. **Section C:** Q. Nos. 26 to 31 are Short Answer (SA) type questions carrying **3 marks** each.
6. **Section D:** Q. Nos. 32 to 35 are Long Answer (LA) type questions carrying **5 marks** each.
7. **Section E:** Q. Nos. 36 to 38 are case study based questions carrying **4 marks** each. Internal choice is provided in the 2-mark sub-part of each case study.
8. There is no overall choice. However, an internal choice has been provided in 2 questions in Section B (Q22, Q25), 2 in Section C (Q26, Q27), 2 in Section D (Q33, Q34) and 3 in Section E (Q36, Q37, Q38).
9. Draw neat diagrams wherever required. Take **π = 22/7** wherever required, if not stated.
10. Use of calculator is **NOT** allowed. Marks for each question are shown in square brackets on the right.

SECTION A - Multiple Choice and Assertion-Reason Questions (20 Marks)

1. For what value of k, the product of zeroes of the polynomial $kx^2 - 4x - 7$ is 2 ? **[1]**

(A) $-\frac{1}{14}$ **(B)** $-\frac{7}{2}$

(C) $\frac{7}{2}$ **(D)** $-\frac{2}{7}$

2. The larger of two supplementary angles exceeds the smaller by 18 degrees. What is the measure of larger angle ? **[1]**

(A) 81° **(B)** 99°

(C) 36° **(D)** 54°

3. Which of the following equations has 2 as a root ? **[1]**

(A) $x^2 - 4x + 5 = 0$ **(B)** $x^2 + 3x - 12 = 0$

(C) $2x^2 - 7x + 6 = 0$ **(D)** $3x^2 - 6x - 2 = 0$

4. The equation $x + \frac{1}{x} = 3$ $(x \neq 0)$ is expressed as a quadratic equation in the form of $ax^2 + bx + c = 0$. The value of a - b + c is : **[1]**

(A) 5 **(B)** 2

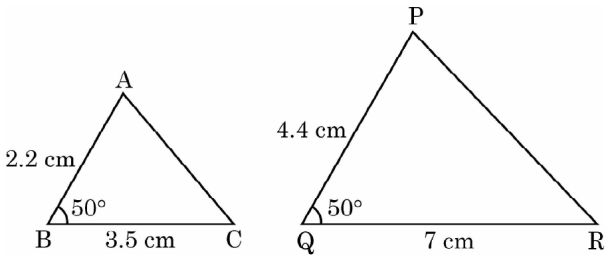
(C) 1 **(D)** -1

5. If 14th term of an A.P. is 4 and its 15th term is zero, then its first term is **[1]**

(A) -48 **(B)** -56

(C) 56 **(D)** 48

6. In the given figure, the criterion of similarity by which ΔABC ~ ΔPQR is : **[1]**



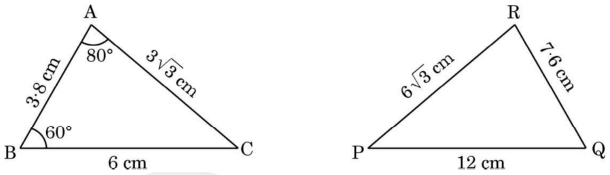
(A) SSA (Side – Side – Angle) Similarity

(B) ASA (Angle – Side – Angle) Similarity

(C) SAS (Side – Angle – Side) Similarity

(D) AA (Angle – Angle) Similarity

7. From the figures given below, which of the following is true about the measure of ∠P ? **[1]**



(A) ∠P = 60°

(B) ∠P = 80°

(C) ∠P = 40°

(D) The measure of ∠P cannot be determined

8. $\frac{2 \tan 30°}{1 - \tan^2 30°}$ is equal to **[1]**

(A) cos 60°

(B) sin 60°

(C) tan 60°

(D) sin 30°

9. If $\sin \theta = \frac{1}{\sqrt{11}}$, then cot θ equals **[1]**

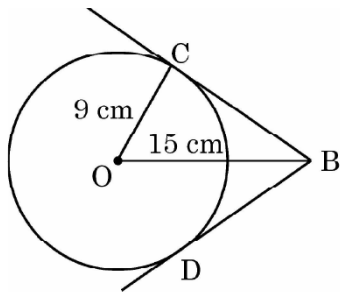
(A) $\frac{\sqrt{11}}{\sqrt{10}}$

(B) $\frac{\sqrt{10}}{\sqrt{11}}$

(C) $\sqrt{10}$

(D) $\sqrt{11}$

10. In the given figure, BC and BD are tangents to the circle with centre O and radius 9 cm. If OB = 15 cm, then the length (BC + BD) is : **[1]**



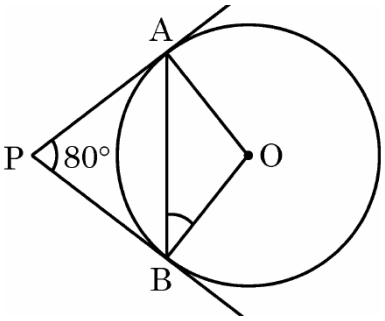
(A) 18 cm

(B) 12 cm

(C) 24 cm

(D) 36 cm

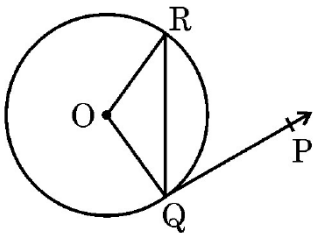
11. In the given figure, tangents PA and PB from a point P to a circle with centre O are inclined to each other at an angle of 80°. ∠ABO is equal to **[1]**



(A) 40° **(B)** 80°

(C) 100° **(D)** 50°

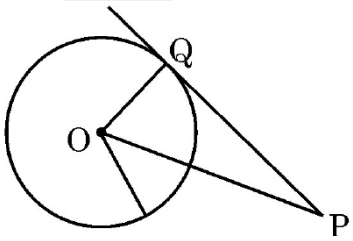
12. A chord QR subtends an angle of 105° at the centre O of the circle. The measure of ∠RQP is **[1]**



(A) $\frac{75°}{2}$ **(B)** $\frac{105°}{2}$

(C) 75° **(D)** 15°

13. PQ is tangent to the circle with centre O such that OP = 2OQ. m∠OPQ is **[1]**



(A) 15° **(B)** 60°

(C) 45° **(D)** 30°

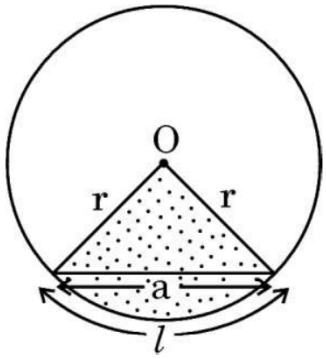
14. The difference of the areas of a minor sector of angle 120° and its corresponding major sector of a circle of radius 21 cm, is **[1]**

(A) 231 cm^2 **(B)** 462 cm^2

(C) 346.5 cm^2 **(D)** 693 cm^2

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15. The perimeter of the shaded region in the given figure is : **[1]**



(A) *l* **(B)** *l* + a
(C) *l* + 2r **(D)** *l* + 2r + a

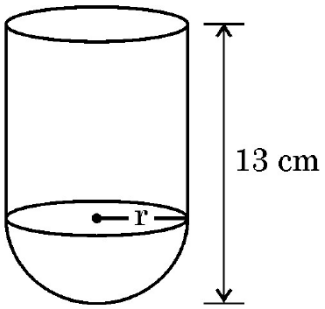
16. Outer surface area of a cylindrical juice glass with radius 7 cm and height 10 cm, is : **[1]**

(A) 440 sq m **(B)** 594 sq m
(C) 748 sq m **(D)** 1540 sq m

17. The radius of a sphere is $\frac{7}{2}$ cm. The volume of the sphere is : **[1]**

(A) $\frac{231}{3}$ cu cm **(B)** $\frac{539}{12}$ cu cm
(C) $\frac{539}{3}$ cu cm **(D)** 154 cu cm

18. A cylinder of radius r is surmounted on a hemisphere of same radius. If total height of the object is 13 cm, then its inner surface area is **[1]**



(A) 2πr(r + 13) **(B)** 13πr
(C) $2\pi(13 + r)^2$ **(D)** 26πr

Assertion-Reason based questions. *Directions* : In Q. Nos. 19 and 20, a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

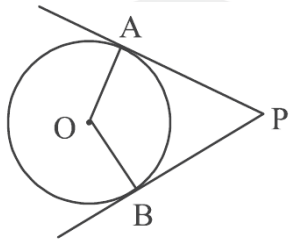
(D) Assertion (A) is false, but Reason (R) is true.

19. **Assertion (A) :** The value of p for which the system of equations $4x + py + 8 = 0$ and $2x + 2y + 2 = 0$ is consistent is 4. **[1]**

Reason (R) : The system of equations $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$ is consistent with infinitely many solutions, if $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$.

20. **Assertion (A) :** If the PA and PB are tangents drawn to a circle with centre O from an external point P, then the quadrilateral OAPB is a cyclic quadrilateral. **[1]**

Reason (R) : In a cyclic quadrilateral, opposite angles are equal.



SECTION B - Very Short Answer Questions (10 Marks)

21. Find the LCM and HCF of 92 and 510, using prime factorisation. **[2]**

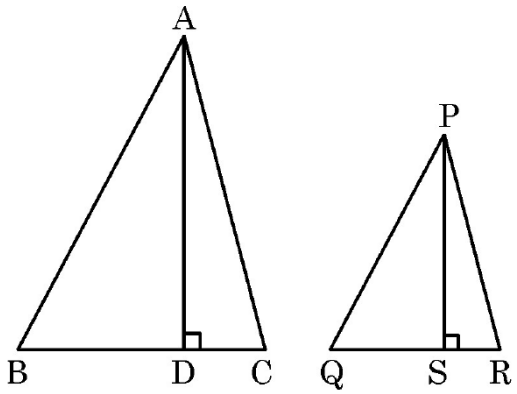
22. (a) Prove that $6 - 4\sqrt{5}$ is an irrational number, given that $\sqrt{5}$ is an irrational number. **[2]**

OR

(b) Show that $11 \times 19 \times 23 + 3 \times 11$ is not a prime number.

23. If H.C.F. of $135x^2$ and $189x^3$ is 108, then find the value of x. **[2]**

24. In the given figure, two triangles ABC and PQR are shown such that $\angle A = \angle P$ and $\angle C = \angle R$. If AD $\perp$ BC and PS $\perp$ QR, then prove that (i) $\Delta ADB \sim \Delta PSQ$ (ii) $AD \times QS = BD \times PS$. **[2]**



25. (a) Find the probability that a number selected at random from the numbers 30, 31, 32, 33, ..., 60 is (i) a prime number (ii) a multiple of 6. **[2]**

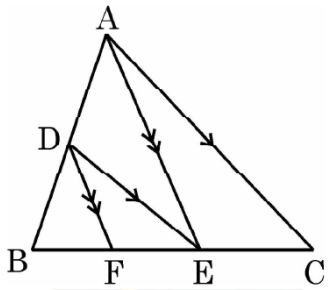
OR

(b) Slips of letters of the word 'BACKGROUND' are put in a bowl and thoroughly mixed. One slip is picked up at random. Find the probability that picked up slip's letter is (i) a vowel (ii) present in the word 'BALL'.

SECTION C - Short Answer Questions (18 Marks)

26. (a) In the given figure, DE ∥ AC and DF ∥ AE **[3]**

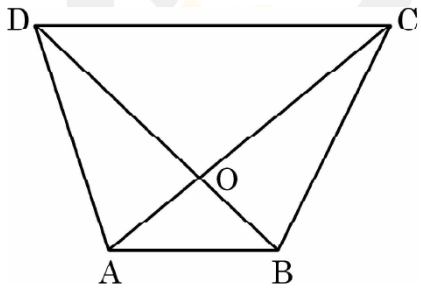
Prove that $\frac{BF}{FE} = \frac{BE}{EC}$



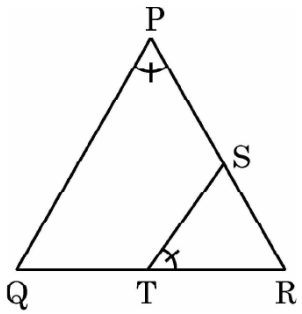
OR

(b) The diagonals of a quadrilateral ABCD intersect each other at the point O such that $\frac{AO}{BO} = \frac{CO}{OD}$

Show that quadrilateral ABCD is a trapezium.



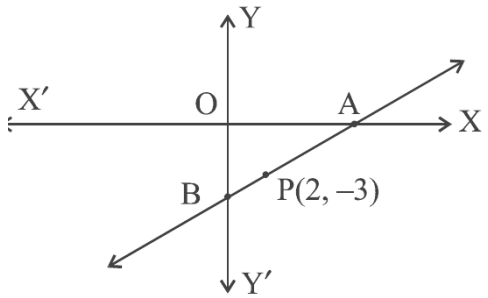
27. (a) S and T are points on sides PR and QR of ΔPQR such that ∠P = ∠RTS. Show that ΔRPQ ~ ΔRTS. **[3]**



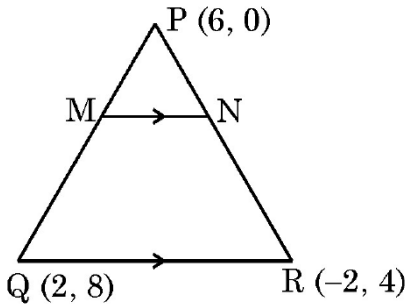
OR

(b) The diagonal BD of a quadrilateral ABCD bisects both ∠B and ∠D. Show that $\frac{AB}{BC} = \frac{AD}{CD}$.

28. The line AB intersects x-axis at A and y-axis at B. The point P(2, -3) lies on AB such that AP : PB = 3 : 1. Find the co-ordinates of A and B. **[3]**



29. Points P(6, 0), Q(2, 8) and R(-2, 4) are vertices of ΔPQR. It is given that MN ∥ QR such that $\frac{PM}{MQ} = \frac{1}{3}$. Using distance formula and ratio formula, show that $\frac{MN}{QR} = \frac{1}{4}$. **[3]**



30. Prove that : $\surd(\frac{\sec A - 1}{\sec A + 1}) + \surd(\frac{\sec A + 1}{\sec A - 1}) = 2 \operatorname{cosec} A$ **[3]**

31. Prove the following trigonometric identity : **[3]**

$(\sin A - \operatorname{cosec} A)(\cos A - \sec A) = \frac{1}{\tan A + \cot A}$

SECTION D - Long Answer Questions (20 Marks)

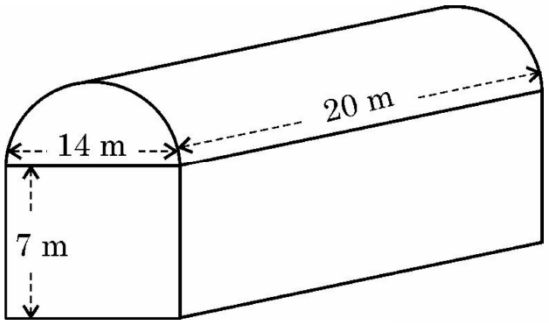
32. A cottage industry produces a certain number of toys in a day. The cost of production of each toy (in rupees) was found to be 55 minus the number of toys produced in a day. On a particular day, the total cost of production was ₹ 750. Find the total number of toys produced on that day. **[5]**

33. (a) ABCD is a rectangle of dimensions 80 cm × 60 cm. Another rectangle PQRS is drawn inside ABCD leaving space of equal width *x* cm along the edges of ABCD. If area PQRS is half of the area ABCD, then find the value of *x*. **[5]**

OR

(b) A train covers a distance of 90 km at a uniform speed. Had the speed been 15 km/h more, it would have taken 30 minutes less for the same journey. Find the original speed of the train.

34. (a) A textile industry runs in a shed. This shed is in the shape of a cuboid surmounted by a half cylinder. If the base of the industry is of dimensions 14 m × 20 m and the height of the cuboidal portion is 7 m, find the volume of air that the industry can hold. Further, suppose the machinery in the industry occupies a total space of 400 m^3. Then, how much space is left in the industry ? **[5]**



OR

(b) From a solid cylinder of height 8 cm and radius 6 cm, a conical cavity of the same height and same radius is carved out. Find the total surface area of the remaining solid. (Take π = 3.14)

35. A survey conducted on 20 families in a locality by a group of students resulted in the following frequency table for the number of family members in a family. **[5]**

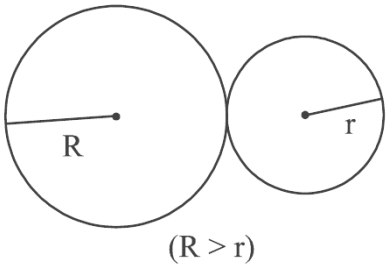
Family size	1 – 3	3 – 5	5 – 7	7 – 9	9 – 11
Number of families	7	8	2	2	1

Determine the mean and mode of the above data.

SECTION E - Case Study Based Questions (12 Marks)

36. Read the case study given below and answer the questions that follow : **[4]**

To keep the lawn green and cool, Sadhna uses water sprinklers which rotate in circular shape and cover a particular area.
The diagram below shows the circular areas covered by two sprinklers :
Two circles touch externally. The sum of their areas is 130 π sq m and the distance between their centres is 14 m.



(R > r)

(36.1) Obtain a quadratic equation involving R and r from above. **[1]**

(36.2) Write a quadratic equation involving only r. **[1]**

(36.3) (a) Find the radius r and the corresponding area irrigated. **[2]**

OR

(36.3) (b) Find the radius R and the corresponding area irrigated. **[2]**

37. Read the case study given below and answer the questions that follow : **[4]**

An injured bird was found on the roof of a building. The building is 15 m high. A fireman was called to rescue the bird. The fireman used an adjustable ladder to reach the roof. He placed the ladder in such a way that the ladder makes an angle of 60° with the ground in order to reach the roof.



(37.1) Find the length of the ladder used by the fireman to reach the roof. **[1]**

(37.2) Find the distance of the point on the ground at which the ladder was fixed from the bottom of the building. **[1]**

(37.3) (a) In order to avoid skidding, the fireman placed the ladder in such a way that the bottom of the ladder touches the base of the wall which is opposite to the building, making an angle of 30° with the ground.
Draw a neat diagram to represent the above situation and hence find the width of the road between the building and the wall. **[2]**

OR

(37.3) (b) In order to avoid skidding, the fireman placed the ladder in such a way that the bottom of the ladder touches the base of the wall which is opposite to the building, making an angle of 30° with the ground.
Find the length of the ladder used by the fireman in this case. **[2]**

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38. Read the case study given below and answer the questions that follow : **[4]**

Heart Rate : The heart rate is one of the 'vital signs' of health in the human body. It measures the number of times per minute that the heart contracts or beats. While a normal heart rate does not guarantee that a person is free of health problems, it is a useful benchmark for identifying a range of health issues.
Thirty women were examined by doctors of AIIMS and the number of heart beats per minute were recorded and summarized as follows :

Number of heart beats per minute	Number of Women
65 – 68	2
68 – 71	4
71 – 74	3
74 – 77	8
77 – 80	7
80 – 83	4
83 – 86	2

(38.1) How many women are having heart beat in the range 68 – 77 ? **[1]**

(38.2) What is the median class of heart beats per minute for these women ? **[1]**

(38.3) (a) Find the modal value of heart beats per minute for these women. **[2]**

OR

(38.3) (b) Find the median value of heart beats per minute for these women. **[2]**

- End of Question Paper -

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All the Best!

You have done the hard work. Now go and show it on the paper.

Exam-day reminders

1	Read the whole paper once and start with the section you are most confident about.
2	Give about 40 minutes to Section A, 20 to Section B, 35 to Section C, 50 to Section D, 25 to Section E, and keep 10 minutes for revision.
3	Write every step. In maths, steps carry marks even if the final answer goes wrong.
4	Draw neat figures in pencil wherever the question needs one, and label them fully.
5	Calculators are not allowed, so practise squares, cubes and square roots till they are quick.
6	Attempt every question. There is internal choice, so pick the one you know best.
7	Solve the 5 papers one by one, from Paper 1 (easy) to Paper 5 (hard), in exam conditions and revise from NCERT after each.

Career Ka Dost
Ask before you decide